OPERA NUMERORUM

p7 Bridge Certificate

BDP Lemma 2 at m=2 | p7 = 631,474,305,334,326,148,720,631 | CERTIFIED

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STATUS: P7_BRIDGE_CERTIFIED

SORRY: 0 | BRIDGE FORMULA VERIFIED | ASCII-ONLY

This certificate formally certifies p7 = 631,474,305,334,326,148,720,631 as the seventh member of S(alpha_0). It verifies primality, S(alpha_0) membership, and the BDP bridge formula at m=2: |191*kappa^2 - p7 - k_bridge*pi| < error_bound_m2.

```
p6 Bridge Certificate SHA-256 (parent):
4f90694c557bb8aec5b425a1b0bd5d355d2095461e439fbaa4ae5e2ddf170ed7
M7 Master Manifest SHA-256 (FROZEN):
5b80b84d1d3d13e216eeecd8155c1edc854d578e7d2dae9c4bc72fcbf7ebe3c9
```

1. p7 Identification and Source

p7 from M4 / S14:

p7 = 631,474,305,334,326,148,720,631 is S14[6] in bound_10_4000.py (Module 4). M4 certifies all 14 members of S(alpha_0) by CF enumeration of alpha_0 at 4010 decimal places. p7 is the seventh exceptional prime, 24 digits.

Rank	Prime in S(alpha_0)	Digits	M4 status
p5	3,993,746,143,633	13	CERTIFIED
p6	3,224,057,731,518,397	16	CERTIFIED
p7	631,474,305,334,326,148,720,631	24	CERTIFIED (this cert)
p8	154,837,899,060,399,532,100,017,991	27	see build_p8_report.py

Source	SHA-256 (first 48 chars)
M4 stdout (bound_10_4000.py)	b810a7a331e47066e3eb4765a5ffdc17c1a56ddbff855a09...

2. Primality Verification

Deterministic Miller-Rabin:

p7 = 631,474,305,334,326,148,720,631 is verified prime by deterministic Miller-Rabin with witnesses {2,3,5,7,11,13,17,19,23,29,31,37}. This witness set is deterministic for all n < 3.317 x 10^24. p7 ~ 6.31 x 10^23 < 3.317 x 10^24, so the test is conclusive.

Test	Value	Result
p7	631,474,305,334,326,148,720,631	PRIME
p7 ~	6.3147 x 10^23 (24 digits)	
Witness set	{2,3,5,7,11,13,17,19,23,29,31,37}	Deterministic
Bound for deterministic	3.317 x 10^24	p7 < bound
Miller-Rabin result	PRIME	CERTIFIED

3. S(alpha_0) Membership

Criterion: ||p7 * alpha_0|| < 1/p7

```
alpha_0 = 299 + pi/10
||p7 * alpha_0|| = 2.2835433400506765011e-25
1/p7 = 1.5835957085705372377e-24
||p7*alpha_0|| < 1/p7: True
```

Quantity	Value	Status
||p7 * alpha_0||	2.28354334005e-25	COMPUTED (100 dps)
1/p7	1.58359570857e-24	COMPUTED
Membership check	True	MEMBER

4. BDP Lemma 2 at m=2

The bridge formula (certified, m=2):

```
|191 * kappa^2 - p7 - k_bridge * pi| < error_bound_m2
191 * kappa^2 = 4480.395610779377769278171
p7 = 631474305334326148720631
k_bridge = -201004514258957634920898
|residual| = 0.00889079107719748
error_bound = 0.0498793602077053
Margin = 0.0409885691305078
PASS: 0.00889079 < 0.04987936
```

Error bound formula at m=2:

error_bound = (m/8)/(2 ln p7) + 1/(2m ln 191)

Term	Formula	Value
Term 1	(2/8)/(2 ln p7)	0.002280924326
Term 2	1/(2*2*ln 191)	0.04759843588
Error bound	Term 1 + Term 2	0.04987936021

bdp2_p7.py stdout SHA-256:

`bff59b704343cfaa30adba5951bf5fef7448edd3828728aa359735e362a7eee6`

5. m-Sequence Decay and Boundary Picture

The BDP bridge exponent m is dropping across the exceptional prime sequence. At p5 the formula required m=16; at p6 it dropped to m=3; at p7 it dropped further to m=2. This decay pattern is the key observation of the P vs NP Tower and the BDP boundary study.

Prime	Digits	m (bridge)	Residual	Error bound	Status
p5 = 3,993,746,143,633	13	16	~0.000285	0.040413	CERTIFIED
p6 = 3,224,057,731,518,397	16	3	0.010112	0.036983	CERTIFIED
p7 = 631,474,305,334,326,148,720,631	24	2	0.008890	0.049879	CERTIFIED
p8 = 154,837,899,060,399,532,100,017,991	27	63	0.022281	0.066805	SEE REPORT

Note: p7 is the last prime certified as both an S(alpha_0) member AND satisfying the bridge formula at a small m (m=2). p8 passes the bridge at m=63 but fails the S(alpha_0) membership criterion: ||p8*alpha_0|| = 0.121 >> 1/p8 = 6.46e-27. See build_p8_report.py for the full boundary analysis.

6. Causal Chain and Status

Chain diagram:

```
M2 (print_kappa.c) kappa = 4.8433014197780389 [SHA 3716c7db...]
|
v
M4 (bound_10_4000.py) S14[6] = 631,474,305,334,326,148,720,631 [SHA b810a7a3...]
|
v
bdp2_p7.py: |191*kappa^2 - p7 - k*pi| < 0.049879 [CERTIFIED]
|
v
p7 = 631,474,305,334,326,148,720,631 [P7_BRIDGE_CERTIFIED]
```

Node	Claim	Status
M2 (print_kappa.c)	kappa = 4.8433014197780389 (80-bit)	CERTIFIED
M4 (bound_10_4000.py)	S14[6] = 631,474,305,334,326,148,720,631	CERTIFIED
Primality (M-R)	p7 is prime (deterministic witnesses)	CERTIFIED
S(alpha_0) membership	||p7*alpha_0|| < 1/p7	CERTIFIED
BDP Lemma 2 (m=2)	|191*kappa^2 - p7 - k*pi| < 0.049879	CERTIFIED
p7 Bridge Cert	All three checks PASS	P7_BRIDGE_CERTIFIED

OVERALL STATUS: P7_BRIDGE_CERTIFIED

p7 = 631,474,305,334,326,148,720,631 (S14[6], M4)

SORRY: 0 | BRIDGE FORMULA VERIFIED (m=2) | ASCII-ONLY

NOTE: m-sequence decaying 16 -> 3 -> 2. p8 exits S(alpha_0). See p8 boundary report.

`M7 MANIFEST LOCKED: 5b80b84d1d3d13e216eeecd8155c1edc...`

All SHAs live-computed. No fabricated values. ASCII only.